

Kexin Wei

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AI/ML engineer specializing in medical image segmentation and deep learning, with full-stack skills to take research models from prototype to validated, production-ready pipelines.

Experience

Research Engineer <i>Imperial Global Singapore, Imperial College London.</i>	Mar 2025 — Present <i>Singapore</i>
AI Agents: Building an agentic (Claude tool-use, multi-step planning) penetration-testing system for medical cyber-physical systems. Simulation: Translated the MATLAB controller to Python and tuned a real-time closed-loop glucose-insulin patient simulator. Full-Stack: Built a Flask + Vue.js application for real-time simulation and cyberattack visualization. DevOps: Dockerized and deployed the simulation stack with multi-container orchestration.	
Software Team Lead <i>Creative Medtech Solutions Pte.Ltd.</i>	Mar 2024 — Jan 2025 <i>Singapore</i>
AI: Implemented AI-based medical image segmentation, 4D DICOM processing, and US/MRI image registration algorithm. CyberSecurity: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI. Lead: Led a team of 6 engineers: Git workflow, Scrum adoption, GoogleTest integration, and code-review standards.	
Research and Development Engineer <i>Creative Medtech Solutions Pte.Ltd.</i>	Jun 2022 — Mar 2024 <i>Singapore</i>
AI: Developed and validated an AI-based 3D medical image segmentation pipeline for prostate biopsy guidance. Image: Prototyped kidney segmentation solutions using SAM, MedSAM, and OpenCV for potential implementation. Robot: Designed respiratory motion tracking algorithm integrating an A*STAR kidney segmentation CNN, verified in vivo across pig and dog trials. Patent: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.	

Education

M.Eng in Medical Robotics and AI <i>National University of Singapore, Singapore</i>	Aug 2020 — May 2022 <i>GPA: 4.63/5.00</i>
- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training. Code. - TA for ME2143E Motor Characteristics and ME5405 Machine Vision.	
B.Eng in Mechanical Engineering <i>Tongji University, China</i>	Sep 2014 — Jul 2019 <i>GPA: 4.61/5.00</i>

Selected Publications

- K. Wei et al.**, “Glucoprint: CGM-Based Re-identification Across Six Type 1 Diabetes Cohorts” (*first author, in prep.*); co-author on *G-FM* (flow-matching time-series generation) and *HealthLoopQA* (clinical QA benchmark), both *submitted 2026*.

Awards & Honors

AI Mathematics Modeling Scholarship	Hackathon “Runner Up” Prize at bio LLMs hosted by 4Catalyzer (2023) 2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018) 3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)
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Skills

AI/ML	PyTorch, TensorFlow, Deep Learning (CV, CNN, DRL), LLM (Claude, OpenAI, Gemini; LangChain/LangGraph)
Programming	Python, C++, PyTorch, SQL, Docker, Linux, Git, Flask, Vue.js, OpenCV, VTK, MATLAB
CAD/CAE	Solidworks, AutoCAD
Language	English (C1), German (C1), Mandarin (native)